

Keywords and Logic Operators Reference Table

Basic Propositional Logic Keywords

Keyword/Phrase	Logical Operator	Symbol	Meaning	Example
not, it is not the case that	Negation	$\neg p$ or $\sim p$	Opposite truth value	"Abby did not send more than 100 texts" $\rightarrow \neg p$
and, but, also	Conjunction	$p \wedge q$	Both must be true	"Smartphone has 256 MB RAM and 32 GB ROM" $\rightarrow p \wedge q$
or	Disjunction	$p \vee q$	At least one is true	"Smartphone C has more ROM or a higher resolution" $\rightarrow p \vee q$
if...then, implies, only if, whenever	Conditional (Implication)	$p \rightarrow q$	If p is true, then q must be true	" If it rains, then the ground is wet" $\rightarrow p \rightarrow q$
if and only if, iff, necessary and sufficient	Biconditional	$p \leftrightarrow q$	Both have same truth value	"Triangle is equilateral iff all sides are equal" $\rightarrow p \leftrightarrow q$
either...or (exclusive)	Exclusive OR	$p \oplus q$	Exactly one is true	" Either pass or fail (not both)" $\rightarrow p \oplus q$

Conditional Statement Keywords ($p \rightarrow q$)

Hypothesis (p) Indicators:

- "if"
- "when"
- "whenever"
- "given that"
- "suppose that"
- "assume that"

Conclusion (q) Indicators:

- "then"

- "implies"
- "it follows that"
- "therefore"
- "only if" (careful: this indicates the **conclusion**, not hypothesis)

Important Variations:

English Form	Logical Form	Notes
If p, then q	$p \rightarrow q$	Standard form
p only if q	$p \rightarrow q$	"only if" introduces the conclusion
p is sufficient for q	$p \rightarrow q$	p being true is enough for q
q if p	$p \rightarrow q$	Reversed word order, same logic
q whenever p	$p \rightarrow q$	Alternative phrasing
q is necessary for p	$p \rightarrow q$	q must be true for p to be true
p implies q	$p \rightarrow q$	Direct statement

Biconditional Keywords ($p \leftrightarrow q$)

Keyword/Phrase	Symbol	Meaning
if and only if	$p \leftrightarrow q$	Both directions: $(p \rightarrow q) \wedge (q \rightarrow p)$
iff	$p \leftrightarrow q$	Abbreviation of "if and only if"
necessary and sufficient	$p \leftrightarrow q$	p is both necessary and sufficient for q
exactly when	$p \leftrightarrow q$	Same truth values required

Quantifier Keywords (From 1.4–1.5)

English Phrase	Quantifier	Symbol	Meaning
for all, for every, for each, all	Universal	$\forall x$	Property holds for every element
there exists, for some, at least one, some	Existential	$\exists x$	Property holds for at least one element
there exists exactly one, there is a unique	Uniqueness	$\exists!x$	Property holds for exactly one element

Negation Patterns

Basic Negation:

- "not p" $\rightarrow \neg p$
- "it is not the case that p" $\rightarrow \neg p$
- "p is false" $\rightarrow \neg p$

Compound Statement Negations (De Morgan's Laws):

Original Statement	Negation	Equivalent Form
$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
$p \vee q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
$p \rightarrow q$	$\neg(p \rightarrow q)$	$p \wedge \neg q$
$\forall x P(x)$	$\neg(\forall x P(x))$	$\exists x \neg P(x)$
$\exists x P(x)$	$\neg(\exists x P(x))$	$\forall x \neg P(x)$

Context-Specific Keywords

Comparison Statements:

- "more than" $\rightarrow$ Use inequality with appropriate direction

- "at least" $\rightarrow \geq$
- "at most" $\rightarrow \leq$
- "exactly" $\rightarrow =$
- "less than" $\rightarrow <$
- "greater than" $\rightarrow >$

Complex Conditionals:

- "if...and only if" $\rightarrow$ Biconditional ($\leftrightarrow$)
 - "if p then q, and if q then p" $\rightarrow$ Biconditional ($\leftrightarrow$)
 - "p unless q" $\rightarrow \neg q \rightarrow p$ (equivalent to $q \vee p$)
 - "p but not q" $\rightarrow p \wedge \neg q$
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Key Takeaways:

1. **"Only if" is tricky:** "p only if q" means $p \rightarrow q$ (not $q \rightarrow p$)
2. **"Or" is inclusive:** Unless stated as "exclusive or," assume inclusive disjunction ($\vee$)
3. **Order matters in English, not always in logic:** "q if p" = "if p then q" = $p \rightarrow q$
4. **Negation distributes:** Use De Morgan's Laws for negating compound statements
5. **Watch for hidden quantifiers:** "All smartphones" implies $\forall x$, "Some smartphone" implies $\exists x$